

Effect of gap-filling technique and gap location on linear and nonlinear calculations of motion during locomotor activities

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ABSTRACT

Background: Marker occlusion during camera-based movement analysis is common. Different interpolation techniques are available for estimating location of missing marker trajectories.

Research question: What is the effect of gap location and interpolation technique on linear and nonlinear measures for a given kinematic time series?

Methods: Kinematic data were recorded during motor-assisted elliptical training and treadmill walking. Gap-filling techniques (i.e., Cubic, Makima, Autoregressive, Nearest Neighbor, and No Interpolation) and gap locations experimentally applied to each cycle across initially complete time series (Gap 1: local minimum and maximum peaks; Gap 2: maximum peaks; Gap 3: maximum peaks at negative slope; Gap 4: random locations) were examined during linear (Maxima and Minima joint angles) and nonlinear [maximum Lyapunov exponent (LyE)] measures.

Results: Gap-filling technique and gap location influenced values calculated for linear and nonlinear measures of joint motions. When referenced to the gold standard (original data series without gaps), across all joints studied the average % error of Maxima and Minima joint angles and LyE % error were lower when applying Cubic, Makima, Autoregressive, and Nearest Neighbor techniques compared to No Interpolation ($p < 0.0001$). The % error of Maxima joint angles was lower for Gaps 1, 3, and 4 compared to Gap 2 ($p = 0.0003$), while % error of Minima joint angles was lower for Gaps 2 and 3, compared to Gaps 1 and 4 ($p < 0.0001$). An interaction between gap-filling technique and gap location was identified for LyE % error, in which Gap 4 % error was significantly greater during No Interpolation compared to other gap-filling techniques ($p < 0.0001$).

Significance: Findings can guide selection of appropriate techniques to manage missing kinematic data points in camera-based motion analysis time series. Gap-filling techniques significantly reduced error in calculating select linear and nonlinear measures of variability, with Cubic most consistently resulting in the greatest reduction in error.

1. Introduction

While preferably avoided, marker occlusion during camera-based motion analysis frequently occurs and can impede data quality and subsequent interpretation of linear and nonlinear measures used to describe human joint kinematics. If only a subset of recorded cycles is required for linear analysis of movement (e.g., average peak hip flexion angle during 10 walking cycles), then researchers may screen data and select only cycles with non-disrupted marker trajectories. However, selectively eliminating cycles potentially alters the magnitude and

variability of linear measures because factor(s) contributing to marker occlusion (e.g., variations in cane placement) may also contribute to systematic alterations of variables studied (e.g., hip flexion Maxima). Selectively eliminating strides also limits the use of nonlinear tools that require long series of contiguous cycles.

To overcome these challenges, researchers frequently use mathematical interpolation routines to estimate the location of missing marker trajectory data points. While many interpolation tools exist [1–5], the impact of using varying tools on the validity of linear and nonlinear measures has not been systematically studied. Accurately representing

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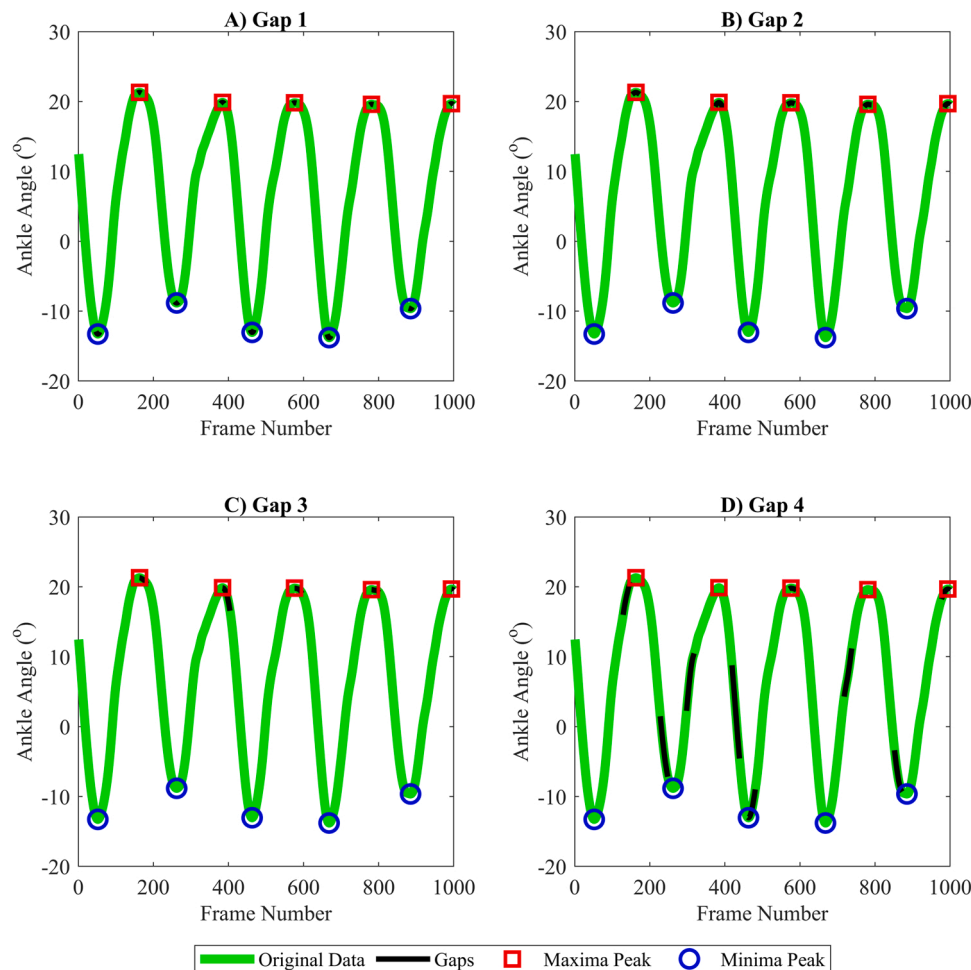


Fig. 1. Exemplar time-series' section on the sagittal plane displaying original data (green line), location of gaps (region covered by the solid black line), Maxima peak (red square), and Minima peak (blue circle). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

joint kinematics is essential not only for surgical/therapeutic decision-making [6], but also when using data to guide wearable robotic assistance [7] or modeling musculoskeletal movement [8].

The current research examined the influence of applying common gap-filling techniques to differing gap locations on values calculated for linear (Maxima, Minima) and nonlinear measures of variability [Lyapunov exponent (LyE)] [9] arising during two cyclic locomotor activities. While linear measures of variability are indicative of “how much” variability is present, nonlinear measures can reveal “organization/disorganization” (non-linear) of movement patterns. Optimal variability is neither too constrained (i.e., lacking any variability across repetitions) nor too variable (i.e., overly disorganized and random). Five approaches to managing gaps (No Interpolation, Cubic, Makima, Autoregressive, and Nearest Neighbor) were applied to four simulated gap types (three associated with Maxima and Minima joint angles, one applied to random locations) that were created in pre-recorded kinematic data to discover each technique's effect on measures of linear and nonlinear kinematic variability. We hypothesized that using Cubic, Makima, Autoregressive, and Nearest Neighbor techniques to interpolate missing data points would result in lower error in select linear (Maxima, Minima) and nonlinear (LyE) measure calculations compared to applying no interpolation. Gaps applied in random locations were expected to lead to more chaotic signals, and thus a higher LyE, compared to gaps applied in predictable locations.

2. Methods

2.1. Participant

A 27-year-old male (weight: 70.2 kg, height: 1.83 m) without known disabilities provided informed consent to participate in a broader study evaluating kinematic and electromyographic demands associated with motor-assisted elliptical training and walking. All consent and testing occurred in Madonna Rehabilitation Hospitals' Institute for Rehabilitation Science and Engineering.

2.2. Instrumentation

The ICARE E872 (SportsArt, Mukilteo, WA, USA) was used during the motor-assisted elliptical condition. ICARE's training parameters (motor-assistance, training speed, step length, and external bodyweight support) can be adjusted to users' capabilities [10–16]. Walking trials were completed on a T655MS medical-grade treadmill (SportsArt, Mukilteo, WA, USA; walking surface 158 × 56 cm).

Kinematic data were recorded (120 Hz) using 12.7-mm reflective markers (B&L Engineering, Santa Ana, CA, USA) and the Qualisys Motion Capture System (Qualisys AB, Gothenburg, Sweden) including 12 Oqus Series-3 cameras. Qualisys Track Manager Software was used for initial data processing. Kinematic data were processed using Visual3D Professional software (CA-Motion, Inc., Germantown, MD). Variability measures were calculated using custom MATLAB programs (The MathWorks, Inc., Natick, MA).

Table 1

Average Maxima (Max) and Minima (Min) joint angles (degrees) for the original Gold Standard data set and for each gap-filling technique.

Plane	Gap	Gold Standard		Cubic		Makima		Autoregressive		Nearest Neighbor		No Interpolation	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Sagittal	Gap 1	-1.84	35.32	-1.58	35.29	-0.95	34.91	-1.74	35.12	-0.25	34.5	0.66	33.04
	Gap 2	-1.84	35.32	-1.84	35.16	-1.84	34.03	-1.84	33.83	-1.84	32.75	-1.84	31.86
	Gap 3	-1.84	35.32	-1.84	35.33	-1.84	35.27	-1.84	35.29	-1.84	35.27	-1.84	33.45
	Gap 4	-1.84	35.32	-1.43	35.27	-1.15	35.09	-1.18	35.09	-1.12	34.98	-0.35	33.66
Frontal	Gap 1	-4.19	2.15	-4.07	2.05	-3.96	1.94	-4.03	2.00	-3.88	1.85	-3.87	1.85
	Gap 2	-4.19	2.15	-4.19	1.85	-4.19	1.66	-4.19	1.68	-4.19	1.53	-4.19	1.53
	Gap 3	-4.19	2.15	-4.19	2.16	-4.19	2.06	-4.19	2.05	-4.19	2.05	-4.19	2.04
	Gap 4	-4.19	2.15	-4.16	2.11	-4.01	2.04	-4.02	2.03	-3.98	2.02	-3.98	2.01
Transverse	Gap 1	-22.58	-11.76	-22.46	-11.83	-22.20	-12.17	-22.32	-12.15	-22.07	-12.39	-22.06	-12.39
	Gap 2	-22.58	-11.76	-22.57	-12.16	-22.57	-12.84	-22.57	-12.98	-22.57	-13.17	-22.57	-13.18
	Gap 3	-22.58	-11.76	-22.66	-11.62	-22.57	-11.88	-22.57	-11.89	-22.57	-11.86	-22.57	-11.91
	Gap 4	-22.58	-11.76	-22.61	-11.97	-22.43	-12.14	-22.42	-12.17	-22.40	-12.22	-22.36	-12.22

2.3. Procedures

As this is a secondary analysis of a single participant's data from a larger data set, the methods have been reported previously [17]. In brief, following identification of self-selected ICARE and treadmill settings (e. g., speed), the participant returned the next day and reflective markers were placed on his body according to the 6 degrees-of-freedom/CAST full-body marker set guidelines [18]. A static calibration trial was recorded to define the subsequent biomechanical model used for analysis. Markers were placed on the lateral anterior and posterior borders of ICARE's pedals to track location and identify cycle timing.

This study used data recorded while the participant trained in the forward direction for 3-minutes with ICARE's motor assisting and then while treadmill walking (0% incline) with hands unsupported. While using the motor-assisted elliptical, participants were instructed to "let the ICARE motor assist you through the motion" and to "not try to override and pedal faster than the motor setting." The motor provided sufficient torque to move the pedals at the self-selected speed.

2.4. Data analysis

Kinematic data recorded during ICARE training and treadmill walking were analyzed. To ascertain the influence of gap-filling technique and gap location on linear (Maxima, Minima) and nonlinear (i.e., maximum LyE) measures, 3500 consecutive data points (~29.2 s) of data recorded during the final minute were exported.

Each marker's three-dimensional kinematics were calculated using Visual3D™ software (C-Motion, Inc., Rockville, MD, USA). Marker trajectories were filtered (fourth-order Butterworth low-pass filter, 6 Hz cut-off frequency). Local coordinate systems for the pelvis, thighs, legs, feet, and arms were derived from the standing calibration trial. Sagittal, frontal and transverse planes orientations of the pelvis and thigh segments were calculated and expressed relative to vertical. Hip was calculated as the motion of thigh relative to pelvis, knee as leg relative to thigh, and ankle as foot relative to leg. Shoulder angles were expressed as humerus relative to trunk.

Five gap-filling techniques were studied. **Cubic** spline involved using a third-degree polynomial specified in Hermite form (i.e., its values and first derivatives at the endpoints were based on the corresponding domain interval) [2,19,20]. **Makima** cubic Hermite, a modified version of the Akima formula that is used to avoid over- and under-estimation errors, required at least 2 points and the interpolated value at a query point was based on the polynomials' piecewise function with a degree that did not exceed three [3,21]. **Autoregressive** specified that the output variable depended linearly on its previous values and a stochastic (an imperfectly predictable) term [22,23]. **Nearest Neighbor** also required at least 2 points, and the interpolated value at a query point was the value at the nearest sample grid point [4,5,24–26]. **No Interpolation** reflected not applying any interpolation technique to data (i.e.,

gaps not filled).

Initially, twelve concurrent sagittal, frontal and transverse planes kinematic time-series (bilateral shoulder, elbow, hip, thigh, knee, and ankle) with no missing data points (the "Gold Standard") were exported for ICARE and treadmill walking. Then, mean Maxima and Minima and their respective standard deviations were calculated and the Wolf algorithm [27] was applied to each time-series that contained no gaps to determine the maximum LyE. The LyE measures separation of infinitesimally close trajectories of a dynamical system in a particular phase space dimension [27].

Then, four gap types were generated for each joint angle within each plane of motion with a customized MATLAB script. For each time-series, gaps were created by deleting data of the same length (0.167 s) within each cycle in the following predetermined locations within each plane of motion:

- **Gap 1:** 10 frames with Not a Number (NaN; i.e., no data value present) in the angles' time-series centered at local minimum and maximum peaks for a total of 20 frames (Fig. 1A).
- **Gap 2:** 20 frames with NaN in the angles' time-series centered at the maximum local peaks (Fig. 1B).
- **Gap 3:** 20 frames with NaN in the time-series of the angles starting at the maximum local peaks with the negative slope (Fig. 1C).
- **Gap 4:** 20 frames with NaN in the time-series of the angles placed at random locations between each local minimum and maximum peaks (Fig. 1D).

The linear (Maxima and Minima) and nonlinear (LyE) measures were calculated for each gap-imposed time-series.

2.5. Statistical analysis

Descriptive statistics were performed using MATLAB. Assumptions of normality were examined for each dependent variable using the Shapiro-Wilk test. If normality assumptions were accepted, then a two-way analysis of variance (5×4 ANOVA) compared the effect of the five gap-filling techniques (No Interpolation, Cubic, Makima, Autoregressive, Nearest Neighbor) and four gap locations (Gaps 1–4) on percentage (%) error [i.e., Absolute value (Gold Standard Value – Gap-Filling Technique Value) / Gold Standard Value] calculated for linear and nonlinear measures for each plane of motion. Tukey post-hoc tests were conducted for pairwise comparisons. If screening revealed normality assumptions were violated, then data were transformed to ranks, and the Friedman repeated-measures ANOVA on ranks identified significant main effects across conditions. Pairwise multiple comparisons were then performed on the ranked data using the Tukey Test. Alpha levels were 0.05.

Table 2

Average Joint Angle % Error [Absolute value (Gold Standard Value – Gap-Filling Technique Value) / Gold Standard Value] for each Gap Location and Gap-Filling Technique.

Plane		Gap	Cubic	Makima	Autoregressive	Nearest Neighbor	No Interpolation	Row mean
Sagittal	MAXIMA	Gap 1	0.30	1.8	2.0	3.3	8.5	3.2
		Gap 2	0.73	5.2	7.1	9.7	13	7.1
		Gap 3	0.90	0.16	0.15	0.17	6.9	1.7
		Gap 4	0.63	0.86	1.3	1.5	6.0	2.1
		Column mean	0.64	2.0	2.6	3.7	8.6	3.5
	MINIMA	Gap 1	2.0	7.0	2.5	13	20	8.8
		Gap 2	0.0	0.0	0.0	0.0	0.0	0.0
		Gap 3	0.0	0.0	0.0	0.0	0.0	0.0
		Gap 4	3.3	5.4	5.5	5.9	13	6.6
		Column mean	1.3	3.1	2.0	4.7	8.3	3.9
Frontal	MAXIMA	Gap 1	3.4	8.4	7.2	12.0	12.0	8.6
		Gap 2	12.0	18.0	19.0	22.0	22.0	19.0
		Gap 3	1.6	2.7	3.2	2.9	3.7	2.8
		Gap 4	3.0	3.9	4.0	4.4	4.5	3.9
		Column mean	4.9	8.3	8.3	10.0	11.0	8.5
	MINIMA	Gap 1	1.8	5.9	3.4	9.3	9.3	5.9
		Gap 2	0.0	0.0	0.0	0.0	0.0	0.0
		Gap 3	0.0	0.0	0.0	0.0	0.0	0.0
		Gap 4	3.4	5.2	5.5	6.5	6.5	5.4
		Column mean	1.3	2.8	2.3	4.0	4.0	2.8
Transverse	MAXIMA	Gap 1	2.6	6.2	5.2	8.9	8.9	6.4
		Gap 2	9.3	15.0	17.0	20.0	20.0	16.0
		Gap 3	1.9	1.1	1.7	1.2	2.1	1.6
		Gap 4	2.8	4.9	5.3	5.8	5.9	4.9
		Column mean	4.1	6.9	7.3	8.9	9.2	7.3
	MINIMA	Gap 1	2.8	4.9	3.5	6.4	6.4	4.8
		Gap 2	0.0	0.0	0.0	0.0	0.0	0.0
		Gap 3	0.4	0.5	0.5	0.5	0.5	0.5
		Gap 4	0.8	1.3	1.4	1.5	2.1	1.4
		Column mean	1.0	1.7	1.4	2.1	2.2	1.7
Sagittal	MAXIMA: Main Effect of Gap-Filling Technique: Cubic, Makima, Autoregressive, Nearest Neighbor < No Interpolation (p < 0.0001). Main Effect of Gap-Location: Gap 1, Gap 3, Gap 4 < Gap 2 (p = 0.0003). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.8834). MINIMA: Main Effect of Gap-Filling Technique: Cubic, Makima, Autoregressive, Nearest Neighbor < No Interpolation (p < 0.0001). Main Effect of Gap-Location: Gap 2, Gap 3 < Gap 1, Gap 4 (p < 0.0001). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.5334).							
Frontal	MAXIMA: Main Effect of Gap-Filling Technique: Not significant (p = 0.1558). Main Effect of Gap-Location: Gap 3, Gap 4, Gap 1 < Gap 2 (p < 0.0001). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.9855). MINIMA: Main Effect of Gap-Filling Technique: Not significant (p = 0.4105). Main Effect of Gap-Location: Gap 2, Gap 3 < Gap 1, Gap 4 (p < 0.0001). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.9086).							
Transverse	MAXIMA: Main Effect of Gap-Filling Technique: Not significant (p = 0.0810). Main Effect of Gap-Location: Gap 3, Gap 4, Gap 1 < Gap 2 (p < 0.0001), Gap 3 < Gap 1 (p < 0.0001). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.9276). MINIMA: Main Effect of Gap-Filling Technique: Not significant (p = 0.3496). Main Effect of Gap-Location: Gap 2, Gap 3, Gap 4 < Gap 1 (p < 0.0001). Interaction Effect of Gap-Filling Technique x Gap-Location: Not significant (p = 0.7844).							

3. Results

Gap-filling technique and gap location influenced values calculated for linear and nonlinear measures of joint motions recorded during the locomotor activities.

3.1. Linear measures

Average Maxima and Minima joint angles for the original Gold Standard data set and when applying each gap-filling technique are displayed in Table 1.

3.1.1. Impact of gap-filling technique on linear measures (Table 2)

In the sagittal plane, the % error varied significantly across gap-filling techniques for mean Maxima joint angles [$F(4, 220) = 7.8$,

$p < 0.0001$] and Minima joint angles [$F(4, 220) = 11$, $p < 0.0001$]. In the frontal plane, no significant differences were identified across gap-filling techniques in the % error for mean Maxima [$F(4, 220) = 1.7$, $p = 0.1558$] nor Minima joint angles [$F(4, 220) = 1.0$, $p = 0.4105$]. In the transverse plane, no significant differences were identified across gap-filling techniques in the % error for mean Maxima [$F(4, 220) = 2.1$, $p = 0.0810$] nor Minima joint angles [$F(4, 220) = 1.1$, $p = 0.3496$]. (see Table 2)

3.1.2. Influence of gap location on linear measures (Table 2)

In the sagittal plane, % error varied significantly across gap locations for mean Maxima [$F(3, 220) = 6.5$, $p = 0.0003$] and Minima joint angles [$F(3, 220) = 9.6$, $p = 0.0304$]. In the frontal plane, % error varied significantly across gap locations for mean Maxima [$F(3, 220) = 21$, $p < 0.0001$] and Minima joint angles [$F(3, 220) = 10$, $p < 0.0001$]. In

Table 3

Average LyE (bit/s) calculated for the original Gold Standard data set and for each gap-filling technique.

Plane	Gap	Gold Standard	Cubic	Makima	Autoregressive	Nearest Neighbor	No Interpolation
Sagittal	Gap 1	1.02	0.93	1.17	0.88	0.93	1.25
	Gap 2	1.02	1.04	1.04	1.25	1.15	1.18
	Gap 3	1.02	1.00	1.10	1.06	1.64	2.02
	Gap 4	1.02	0.92	1.11	1.32	3.22	8.42
Frontal	Gap 1	2.03	1.95	1.98	1.98	2.17	2.92
	Gap 2	2.03	2.16	2.24	2.52	2.36	2.65
	Gap 3	2.03	2.26	2.15	2.28	2.57	3.76
	Gap 4	2.03	1.96	2.42	2.26	3.03	6.63
Transverse	Gap 1	3.39	2.63	3.13	3.11	3.13	3.23
	Gap 2	3.39	3.20	3.40	3.91	3.48	4.05
	Gap 3	3.39	2.78	3.46	2.98	4.83	5.25
	Gap 4	3.39	2.95	3.25	3.29	4.51	8.57

Table 4

Average LyE % Error [Absolute value (Gold Standard Value – Gap-Filling Technique Value) / Gold Standard Value] for each Gap-Filling Technique and Gap Location.

Plane	Gap	Cubic	Makima	Autoregressive	Nearest Neighbor	No Interpolation	Row means
Sagittal	Gap 1	20	31	22	17	26	23
	Gap 2	18	24	38	34	45	32
	Gap 3	17	32	24	88	145	61
	Gap 4	14	25	61	301	1025	285
	Column means	17	28	36	110	310	100
Frontal	Gap 1	11	30	15	24	60	28.0
	Gap 2	25	36	41	39	51	38.0
	Gap 3	31	27	33	44	118	51.0
	Gap 4	36	54	38	74	341	109.0
	Column means	26	37	32	45	142	56.0
Transverse	Gap 1	24	23	24	31	31	27
	Gap 2	31	25	39	33	37	33
	Gap 3	31	26	35	126	120	68
	Gap 4	27	49	55	102	373	121
	Column means	28	31	39	73	140	62
Sagittal	Main Effect of Gap-Filling Technique: Cubic, Makima, Autoregressive, Nearest Neighbor < No Interpolation ($P < 0.0001$)						
	Main Effect of Gap-Location: Gap 1, Gap 2, Gap 3 < Gap 4 ($p < 0.0001$).						
	Interaction Effect of Gap-Filling Technique x Gap-Location: Significant ($p < 0.0001$).						
	@ Gap 4: Cubic, Makima, Nearest Neighbor, Autoregressive < No Interpolation ($p < 0.0001$).						
	@ No Interpolation: Gap 1, Gap 2, Gap 3 < Gap 4 ($p < 0.0001$).						
Frontal	Main Effect of Gap-Filling Technique: Cubic, Autoregressive, Makima, Nearest Neighbor < No Interpolation ($P < 0.0001$)						
	Main Effect of Gap-Location: Gap 1, Gap 2, Gap 3 < Gap 4 ($p = 0.0002$).						
	Interaction Effect of Gap-Filling Technique x Gap-Location: Significant ($p = 0.0001$).						
	@ Gap 4: Cubic, Autoregressive, Makima, Nearest Neighbor < No Interpolation ($p < 0.0001$).						
	@ No Interpolation: Gap 1, Gap 2, Gap 3 < Gap 4 ($p < 0.0001$).						
Transverse	Main Effect of Gap-Filling Technique: Cubic, Makima, Autoregressive, < No Interpolation ($P = 0.0002$)						
	Main Effect of Gap-Location: Gap 1, Gap 2 < Gap 4 ($p = 0.0004$).						
	Interaction Effect of Gap-Filling Technique x Gap-Location: Significant ($p = 0.0003$).						
	@ Gap 4: Autoregressive, Cubic, Makima, Autoregressive, Nearest Neighbor < No Interpolation ($p < 0.0001$).						
	@ No Interpolation: Gap 1, Gap 2, Gap 3 < Gap 4 ($p < 0.0001$).						

the transverse plane, the % error varied significantly across gap locations for mean Maxima [$F(3, 220) = 26, p < 0.0001$] and Minima joint angles [$F(3, 220) = 25, p < 0.0001$]. No interactions were observed between gap-filling technique and gap location on linear measures within any plane.

3.2. Nonlinear measures

Average LyE values calculated for the original Gold Standard data set, when applying No Interpolation to gaps, and when applying each interpolation method are displayed in Table 3.

3.2.1. Impact of gap-filling technique on nonlinear measures (Table 4)

The % error calculated for mean LyE varied significantly across gap-filling techniques in sagittal [$F(4, 220) = 6.3, p < 0.0001$], frontal [$F(4, 220) = 10, p < 0.0001$], and transverse planes [$F(4, 220) = 5.9, p = 0.0002$]. (see Table 4)

3.2.2. Influence of gap location on nonlinear measures (Table 4; Fig. 2)

The % error calculated for mean LyE varied significantly across gap

locations in sagittal [$F(3, 220) = 8.0, p < 0.0001$], frontal [$F(3, 220) = 7.0, p = 0.0002$], and transverse planes [$F(3, 220) = 6.3, p = 0.0004$]. Significant interactions between gap-filling technique and gap location were identified for the % error calculated for mean LyE in the sagittal [$F(12, 220) = 4.4, p < 0.0001$], frontal [$F(12, 220) = 3.4, p = 0.0001$], and transverse planes [$F(12, 220) = 3.2, p = 0.0003$]. (see Fig. 2)

4. Discussion

The need for gap-filling is not uncommon in human movement analysis given that equipment, body parts, and assistants may inadvertently occlude markers [11,28–30]. The current research identified that applying each interpolation technique (i.e., Cubic, Makima, Autoregressive, and Nearest Neighbor) to any of the gaps studied consistently improved accuracy of linear (Maxima and Minima joint angles) and non-linear (LyE) measure calculations compared to No Interpolation. However, subtle differences in calculated linear and non-linear values highlight the importance of understanding the underlying mathematical algorithm used for interpolating data gaps in varying locations (e.g.,

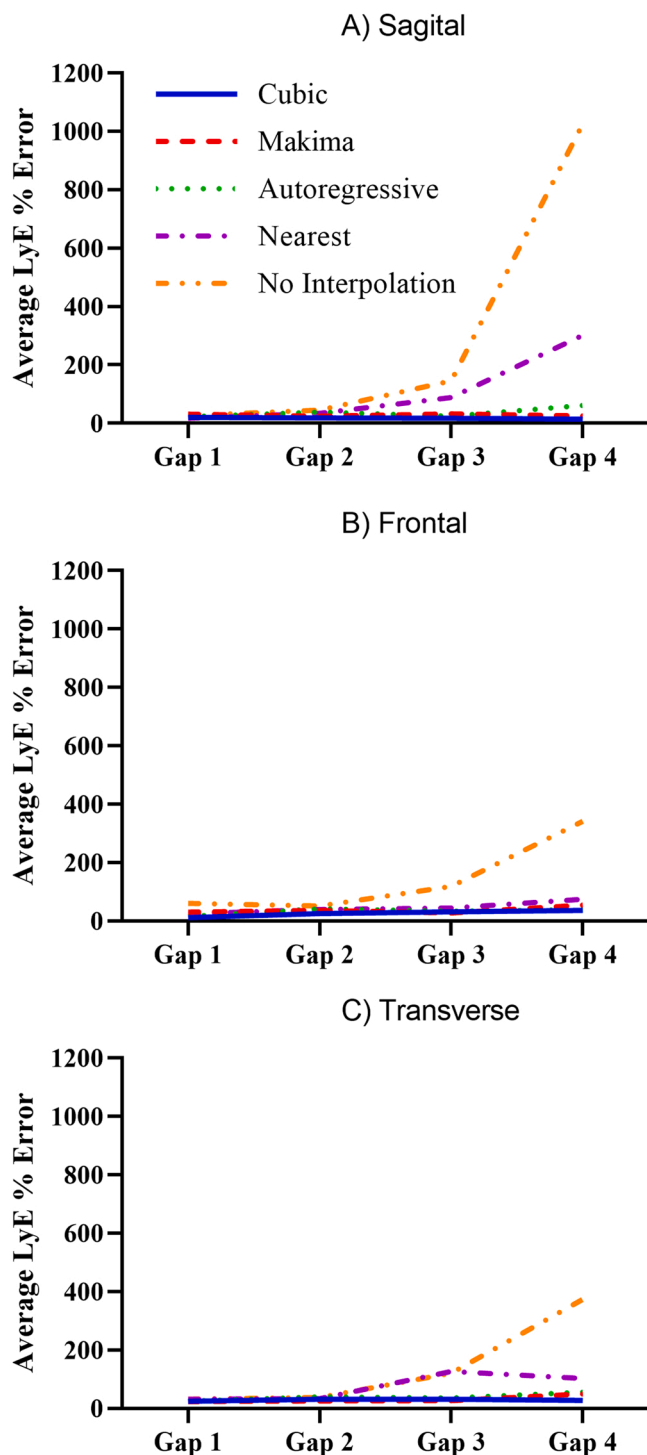


Fig. 2. Interaction between Gap-Filling Technique and Gap Location on LyE % Error. A) Sagittal plane. B) Frontal plane. C) Transverse plane.

zeniths/nadirs vs. random locations).

4.1. Linear measures

Consistent with our hypothesis, errors in linear calculations of Maxima and Minima joint angles were lowered by applying each interpolation technique compared to not interpolating missing data. While not statistically superior, Cubic, on average, produced the most accurate peak joint angle calculations relative to Makima, Nearest Neighbor, and Autoregressive. Cubic's apparent advantage accords with

other studies [1,31] and supports the technique's use to automate gap-filling during inverse kinematic analyses of marker-based motion data [32].

When one-dimensional interpolations are needed (e.g., knee flexion) and Cubic is not available, Makima can be used. It performs cubic interpolations to produce piecewise polynomials with continuous first-order derivatives [3,21]. Makima is reportedly more effective than Cubic at minimizing amplitude of undulations that arise when filling gaps containing abrupt changes between local peaks/valleys and adjacent relatively linear regions of a continuous signal (e.g., Gap 3) [33].

Our data suggest that application of Autoregressive to estimate data points in longer gaps placed symmetrically at the signal's consecutive cycle zeniths (Gap 2) less accurately replicated data points critical for predicting Maxima peak angle. While Autoregressive is well-suited for data extrapolation [34] and for predicting future values [1,35], Gap 2's longer (0.167 s) length and location hindered accurate prediction of missing data points.

Nearest Neighbor is considered computationally easy [22,36]; however, it generated the least accurate estimation of Maxima and Minima joint angles, especially when data points were missing at the signal's zenith and nadir, respectively. Given the technique assigns the value of the closest known (existing) neighbor to the missing value, it is not surprising that the technique blunted calculations of Maxima joint angles for Gaps 1 and 2 as each eliminated data points at, and adjacent to, the consecutive zeniths. Gaps applied in random locations (Gap 4) occasionally occluded data points traversing the zenith or nadir, thus impacting estimated value accuracy. In contrast, the % error associated with calculating Maxima joint angle during Gap 3, which started at each cycle's zenith, was impacted only minimally as the "nearest neighbor" data point preceding the gap closely approximated the zenith's amplitude.

Gap location significantly impacted calculation of linear joint angle calculations (Maxima and Minima), and the contributing factor appeared to be how many data points were missing adjacent to respective zeniths or nadirs. For example, when calculating Maxima, the average percent error was over 50% smaller when occlusions included a briefer portion of each cycle's zenith (generally less than <0.084 s for Gaps 1, 3 and 4) compared to extended gap duration (0.167 s) that spanned Gap 2's zenith. Similarly, when calculating Minima, the 0.0% average percent errors calculated for Gaps 2 and 3 (which only impacted the zenith) were expectedly less than Gap 1 (8.8%) and Gap 4 (6.6%) that included missing nadir data.

4.2. Nonlinear measures

In support of our hypothesis, not interpolating kinematic data with missing data points generated the highest LyE % error across all gaps compared to applying any interpolation technique studied. Additionally, when gaps were introduced randomly (Gap 4), the LyE % error was notably larger when No Interpolation was applied. Given that nonlinear analysis is often used to imply/interpret the functional adaptability/variability of a system [37], the high LyE % error values that arose when No Interpolation was applied, particularly when gaps were in random locations, is concerning. Compared to the same data series recorded without gaps, a data series with randomly located missing data points could be mischaracterized as more chaotic or disorganized [27,38,39], leading to misinterpretation of an intervention's effect or a pathology's progression.

The gap-filling technique by gap location interaction observed further supported our hypothesis. Compared to No Interpolation, LyE % error in the sagittal plane was substantially reduced by applying Cubic (~73-fold reduction), Makima (~41-fold), Autoregressive (17-fold), and Nearest Neighbor (3-fold) to the data series that excluded random data points (Gap 4). Interestingly, when the missing data points were placed consistently at the signal's minimum and maximum peaks (Gap 1), the decision to not interpolate the data rendered LyE error that paralleled

the errors observed for the other gap-filling approaches. While 20 data points were eliminated from each movement cycle across all gaps, Gap 1 eliminated 10 at each cycle's zenith and nadir, possibly mediating impact on the LyE. Collectively these data point to the importance of carefully selecting gap-filling interpolation techniques, particularly when reporting LyE.

4.3. Limitations

The gaps studied do not reflect all missing data point combinations or data gap lengths that might occur across biological signals. Additionally, the effect of each interpolation technique were studied separately versus in combination. Finally, gaps were applied to joint angles versus markers as joint angles are frequently used clinically to describe human motion. Some commercial motion capture software only allows for filling of marker trajectory gaps, thus future research should explore the impact of gap-filling techniques on error calculations arising from occlusion of specific marker trajectories.

5. Conclusion

Clinical gait and biomechanics laboratories often rely on select linear and nonlinear measures of joint kinematics to characterize pathology's or an intervention's impact on locomotion [40]. Missing data points within the time series significantly influenced calculated peak joint angles (Maxima, Minima) and nonlinear measures of movement variability (LyE), and the gap's location (zenith, nadir, slope) and length (number of data points) appeared important. When selecting the optimal technique for filling gaps in joint angle data prior to calculating peak joint angles, we recommend using the Cubic method. In 23 of the 24 comparisons, application of the Cubic method resulted in the lowest % error in peak joint angles when compared to the other gap-filling techniques. In contrast, the least accurate method was Nearest Neighbor. Similarly, when selecting the optimal gap-filling technique prior to non-linear calculations, the Cubic Method resulted in the lowest % error in the majority of comparisons ($n = 7$ of 12), especially in the sagittal and frontal planes. Makima also performed relatively well (lowest % error in 4 of 12 comparisons), particularly in the transverse plane. Our findings can guide processing endeavors of gait-like activities with long series of contiguous strides towards the appropriate techniques that can be used to manage different conditions of missing data points.

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Judith M. Burnfield and Thad W. Buster are the inventors of the patented ICARE. The patented technology has been licensed to SportsArt for commercial distribution. Madonna Rehabilitation Hospitals receives royalties, and a portion of these royalties is shared with inventors. For the remaining authors, no conflicts of interest were declared.

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